

Research Hypotheses

KJV 1611 / KJV 1769 Semantic Drift Study

Purpose

This document defines the null hypothesis and four research hypotheses governing the KJV 1611 / KJV 1769 semantic drift study. The hypotheses connect the study design to the alignment, thresholding, permutation, interpretability, and exclusive corpus components of the workflow.

Null Hypothesis

H0 states that the distribution of semantic distances between aligned KJV 1611 / KJV 1769 text units is not significantly lower than distances produced by randomized text unit pairings under the relevant null model scope.

H1: Canonical Semantic Stability

H1 states that aligned KJV 1611 / KJV 1769 text unit pairs will display significantly lower semantic distance than randomized baseline pairs. This hypothesis is evaluated through Monte Carlo permutation baselines and Stability Gap Index measures.

H2: Presence of Tail Drift

H2 states that, despite high central stability, the shared corpus contains a small upper tail population of text units with unusually high semantic distance relative to robust local thresholds. This hypothesis is evaluated through stratified thresholding, review-candidate counts, strong outlier counts, and upper tail drift metrics.

H3: Structured Textual Change in the Drift Tail

H3 states that flagged high drift units exhibit identifiable linguistic change categories rather than unstructured random variation. This hypothesis is evaluated through HITL annotation, category counts, and enrichment and overrepresentation testing against the inlier background sample.

H4: KJV 1611 Exclusive Corpus Positioning

H4 states that KJV 1611 exclusive material can be positioned within meaningful semantic neighborhoods relative to the shared canon. The current study inventories the exclusive corpus and defers formal H4 nearest neighbor testing to future work because the exclusive material lacks direct KJV 1769 aligned counterparts.

Interpretive Boundary

The hypotheses are designed to support computational and linguistic interpretation. They do not convert embedding distance into doctrinal conclusions without structured human review and appropriate theological or philological analysis.

Study Contract

Component	Study Decision
H0	Aligned distances are not significantly lower than randomized distances.
H1	True aligned pairs are closer than randomized baseline pairs.
H2	A small robustly defined upper tail drift population exists.
H3	Flagged units exhibit identifiable textual change categories.
H4	KJV 1611 exclusive material is inventoried now and analyzed through nearest neighbor methods later.

Expected Output Artifacts

- null_distribution_summary.csv
- drift_labels_by_unit.csv
- category_enrichment_results.csv
- exclusive_dataset_inventory.csv